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Enter password: *****

Welcome to the MariaDB monitor. Commands end with ; or \g.

Your MariaDB connection id is 3

Server version: 12.2.2-MariaDB MariaDB Server

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE Projects;
Query OK, 1 row affected (0.011 sec)

MariaDB [(none)]> USE Projects;
Database changed

MariaDB [Projects]> CREATE TABLE Department(deptNo VARCHAR(10) PRIMARY
KEY,deptName VARCHAR(30), mgrEmpNo INT);
Query OK, 0 rows affected (0.062 sec)

MariaDB [Projects]> CREATE TABLE Employee(empNo VARCHAR(10) PRIMARY KEY,fName
VARCHAR(30),lName VARCHAR(30),address VARCHAR(50), DOB DATE,sex CHAR(1),position
VARCHAR(30),deptNo VARCHAR(10), FOREIGN KEY (deptNo) REFERENCES
Department(deptNo), CONSTRAINT chk_sex CHECK (sex IN('M','F')), CONSTRAINT
chk_position CHECK (position IN('Manager','Team Leader','Analyst','Software
Developer')));
Query OK, 0 rows affected (0.028 sec)

MariaDB [Projects]> CREATE TABLE Project(projNo VARCHAR(10) PRIMARY KEY,
projName VARCHAR(40),deptNo VARCHAR(10), FOREIGN KEY (deptNo) REFERENCES
Department(deptNo));
Query OK, 0 rows affected (0.018 sec)

MariaDB [Projects]> CREATE TABLE WorksOn(empNo VARCHAR(10),projNo
VARCHAR(10),dateWorked DATE,hoursWorked INT,PRIMARY
KEY(empNo,projNo,dateWorked), FOREIGN KEY (empNo) REFERENCES
Employee(empNo),FOREIGN KEY (projNo) REFERENCES Project(projNo), CONSTRAINT
chk_hoursWorked CHECK (hoursWorked >=0 AND hoursWorked <= 40));
Query OK, 0 rows affected (0.022 sec)

MariaDB [Projects]> ALTER TABLE Department ADD CONSTRAINT fk_manager FOREIGN KEY
(mgrEmpNo) REFERENCES Employee(empNo);
Query OK, 0 rows affected (0.059 sec)
Records: 0 Duplicates: 0 Warnings: 0

MariaDB [Projects]> SET FOREIGN_KEY_CHECKS=0;
Query OK, 0 rows affected (0.026 sec)

MariaDB [Projects]> INSERT INTO Department (deptNo, deptName, mgrEmpNo) VALUES
-> ('D1', 'Executive', 'E1'),
-> ('D2', 'Human Resources', 'E2'),
-> ('D3', 'Software Development', 'E3'),
-> ('D4', 'Data Analytics', 'E5'),
-> ('D5', 'Project Management', 'E4'),
-> ('D6', 'Quality Assurance', 'E7'),
-> ('D7', 'IT Support', 'E9'),
-> ('D8', 'Research & Development', 'E6'),
-> ('D9', 'Marketing', 'E8'),
-> ('D10', 'Finance', 'E10');
Query OK, 10 rows affected (0.037 sec)
Records: 10 Duplicates: 0 Warnings: 0

```

MariaDB [Projects]> INSERT INTO Employee (empNo, fName, lName, address, DOB,
sex, position, deptNo) VALUES
-> ('E1', 'Alice', 'Smith', '123 Maple St, London', '1980-05-12', 'F',
'Manager', 'D1'),
-> ('E2', 'Bob', 'Jones', '456 Oak Rd, Manchester', '1985-08-22', 'M',
'Manager', 'D2'),
-> ('E3', 'Charlie', 'Brown', '789 Pine Ave, Bristol', '1988-11-02', 'M',
'Manager', 'D3'),
-> ('E4', 'Diana', 'Prince', '101 Cedar Ln, Leeds', '1990-03-15', 'F',
'Manager', 'D5'),
-> ('E5', 'Evan', 'Wright', '202 Birch Dr, Liverpool', '1992-07-19', 'M',
'Manager', 'D4'),
-> ('E6', 'Fiona', 'Gallagher', '303 Elm St, Newcastle', '1991-01-30', 'F',
'Team Leader', 'D8'),
-> ('E7', 'George', 'Clark', '404 Willow Rd, Sheffield', '1993-09-05', 'M',
'Team Leader', 'D6'),
-> ('E8', 'Hannah', 'Abbott', '505 Ash Blvd, Nottingham', '1994-04-25', 'F',
'Analyst', 'D9'),
-> ('E9', 'Ian', 'Malcolm', '606 Walnut Ter, Leicester', '1987-12-11', 'M',
'Software Developer', 'D7'),
-> ('E10', 'Julia', 'Roberts', '707 Chestnut Ct, Southampton', '1989-06-14',
'F', 'Analyst', 'D10');
Query OK, 10 rows affected (0.034 sec)
Records: 10 Duplicates: 0 Warnings: 0

```

```

MariaDB [Projects]> INSERT INTO Project (projNo, projName, deptNo) VALUES
-> ('SCCS', 'Corporate Restructuring', 'D1'),
-> ('P2', 'Talent Acquisition 2026', 'D2'),
-> ('P3', 'E-Commerce Platform Redesign', 'D3'),
-> ('P4', 'Predictive Market Analytics', 'D4'),
-> ('P5', 'Agile Workflow Optimization', 'D5'),
-> ('P6', 'Automated Regression Testing', 'D6'),
-> ('P7', 'Cloud Infrastructure Migration', 'D7'),
-> ('P8', 'Next-Gen Core Engine R&D', 'D8'),
-> ('P9', 'Global Brand Campaign', 'D9'),
-> ('P10', 'Fiscal Q3 Audit Prep', 'D10');
Query OK, 10 rows affected (0.037 sec)
Records: 10 Duplicates: 0 Warnings: 0

```

```

MariaDB [Projects]> INSERT INTO WorksOn (empNo, projNo, dateWorked, hoursWorked)
VALUES
-> ('E1', 'SCCS', '2026-06-01', 35),
-> ('E2', 'P2', '2026-06-01', 40),
-> ('E3', 'P3', '2026-06-02', 38),
-> ('E4', 'P5', '2026-06-02', 20),
-> ('E5', 'P4', '2026-06-03', 15),
-> ('E6', 'P8', '2026-06-03', 40),
-> ('E7', 'P6', '2026-06-04', 30),
-> ('E8', 'P9', '2026-06-04', 25),
-> ('E9', 'P7', '2026-06-05', 40),
-> ('E10', 'P10', '2026-06-05', 10);
Query OK, 10 rows affected (0.037 sec)
Records: 10 Duplicates: 0 Warnings: 0

```

```

MariaDB [Projects]> SET FOREIGN_KEY_CHECKS = 1;
Query OK, 0 rows affected (0.000 sec)

```

```

MariaDB [Projects]> SELECT * FROM Department;
+-----+-----+-----+
| deptNo | deptName          | mgrEmpNo |
+-----+-----+-----+
| D1     | Executive         | E1       |
| D10    | Finance           | E10      |
| D2     | Human Resources   | E2       |

```

D3	Software Development	E3
D4	Data Analytics	E5
D5	Project Management	E4
D6	Quality Assurance	E7
D7	IT Support	E9
D8	Research & Development	E6
D9	Marketing	E8

10 rows in set (0.013 sec)

MariaDB [Projects]> SELECT * FROM Employee;

empNo	fName	lName	address	DOB	sex
position		deptNo			
E1	Alice	Smith	123 Maple St, London	1980-05-12	F
Manager		D1			
E10	Julia	Roberts	707 Chestnut Ct, Southampton	1989-06-14	F
Analyst		D10			
E2	Bob	Jones	456 Oak Rd, Manchester	1985-08-22	M
Manager		D2			
E3	Charlie	Brown	789 Pine Ave, Bristol	1988-11-02	M
Manager		D3			
E4	Diana	Prince	101 Cedar Ln, Leeds	1990-03-15	F
Manager		D5			
E5	Evan	Wright	202 Birch Dr, Liverpool	1992-07-19	M
Manager		D4			
E6	Fiona	Gallagher	303 Elm St, Newcastle	1991-01-30	F
Team Leader		D8			
E7	George	Clark	404 Willow Rd, Sheffield	1993-09-05	M
Team Leader		D6			
E8	Hannah	Abbott	505 Ash Blvd, Nottingham	1994-04-25	F
Analyst		D9			
E9	Ian	Malcolm	606 Walnut Ter, Leicester	1987-12-11	M
Software Developer		D7			

10 rows in set (0.005 sec)

MariaDB [Projects]> SELECT * FROM Project;

projNo	projName	deptNo
P10	Fiscal Q3 Audit Prep	D10
P2	Talent Acquisition 2026	D2
P3	E-Commerce Platform Redesign	D3
P4	Predictive Market Analytics	D4
P5	Agile Workflow Optimization	D5
P6	Automated Regression Testing	D6
P7	Cloud Infrastructure Migration	D7
P8	Next-Gen Core Engine R&D	D8
P9	Global Brand Campaign	D9
SCCS	Corporate Restructuring	D1

10 rows in set (0.002 sec)

MariaDB [Projects]> SELECT * FROM WorksOn;

empNo	projNo	dateWorked	hoursWorked
E1	SCCS	2026-06-01	35
E10	P10	2026-06-05	10

E2	P2	2026-06-01	40
E3	P3	2026-06-02	38
E4	P5	2026-06-02	20
E5	P4	2026-06-03	15
E6	P8	2026-06-03	40
E7	P6	2026-06-04	30
E8	P9	2026-06-04	25
E9	P7	2026-06-05	40

10 rows in set (0.002 sec)

```
MariaDB [Projects]> CREATE VIEW FemaleManagersProjects AS SELECT
projNo,projName,Project.deptNo, fName,lName FROM Project JOIN Department ON
Project.deptNo=Department.deptNo JOIN Employee ON
Department.mgrEmpNo=Employee.empNo WHERE Employee.sex='F' ORDER BY
Project.projNo;
Query OK, 0 rows affected (0.044 sec)
```

```
MariaDB [Projects]> SELECT * FROM FemaleManagersProjects;
```

projNo	projName	deptNo	fName	lName
P10	Fiscal Q3 Audit Prep	D10	Julia	Roberts
P5	Agile Workflow Optimization	D5	Diana	Prince
P8	Next-Gen Core Engine R&D	D8	Fiona	Gallagher
P9	Global Brand Campaign	D9	Hannah	Abbott
SCCS	Corporate Restructuring	D1	Alice	Smith

5 rows in set (0.003 sec)

```
MariaDB [Projects]> CREATE VIEW EmployeeProjectHours AS SELECT e.empNo, e.fName,
e.lName, p.projName, w.hoursWorked FROM Employee e JOIN WorksOn w ON
e.empNo=w.empNo JOIN Project p ON p.projNo=w.projNo;
Query OK, 0 rows affected (0.034 sec)
```

```
MariaDB [Projects]> SELECT * FROM EmployeeProjectHours;
```

empNo	fName	lName	projName	hoursWorked
E1	Alice	Smith	Corporate Restructuring	35
E10	Julia	Roberts	Fiscal Q3 Audit Prep	10
E2	Bob	Jones	Talent Acquisition 2026	40
E3	Charlie	Brown	E-Commerce Platform Redesign	38
E4	Diana	Prince	Agile Workflow Optimization	20
E5	Evan	Wright	Predictive Market Analytics	15
E6	Fiona	Gallagher	Next-Gen Core Engine R&D	40
E7	George	Clark	Automated Regression Testing	30
E8	Hannah	Abbott	Global Brand Campaign	25
E9	Ian	Malcolm	Cloud Infrastructure Migration	40

10 rows in set (0.038 sec)

```
MariaDB [Projects]> CREATE VIEW EmpProject(empNo,projNo,totalHours) AS SELECT
w.empNo,w.projNo,SUM(hoursWorked) FROM Employee e, Project p, WorksOn w WHERE
e.empNo=w.empNo AND p.projNo=w.projNo GROUP BY w.empNo,w.projNo;
Query OK, 0 rows affected (0.045 sec)
```

```
MariaDB [Projects]> SELECT * FROM EmpProject;
```

+-----+-----+-----+

empNo	projNo	totalHours
E1	SCCS	35
E10	P10	10
E2	P2	40
E3	P3	38
E4	P5	20
E5	P4	15
E6	P8	40
E7	P6	30
E8	P9	25
E9	P7	40

10 rows in set (0.039 sec)

```
MariaDB [Projects]> SELECT projNo FROM EmpProject WHERE projNo='SCCS';
```

projNo
SCCS

1 row in set (0.041 sec)

```
MariaDB [Projects]> SELECT COUNT(projNo) FROM EmpProject WHERE empNo='E1';
```

COUNT(projNo)
1

1 row in set (0.039 sec)

```
MariaDB [Projects]> SELECT empNo, totalHours FROM EmpProject GROUP BY empNo;
```

empNo	totalHours
E1	35
E10	10
E2	40
E3	38
E4	20
E5	15
E6	40
E7	30
E8	25
E9	40

10 rows in set (0.011 sec)

Geneal(7.28)

Answer: A materialized view stores the query result and must be updated whenever the Part table changes and if a new row is inserted or updated so that partCost becomes greater than £1000, the partNo can be added to the view without accessing the base table. However, if an existing row is deleted or its partCost is less than or equal to £1000, the Part table must be checked to see if another contract for the same partNo still has a cost above £1000. If so, the part remains in the view; otherwise, it is removed.